MSCF Probability – Final Exam – August 17, 2018

Name ________________________________

Email ________________________________

Instructions:

- You should complete all questions on this exam by writing out the solutions on paper that will be provided to you. Anything written on the exam paper itself will **not** be graded.
- You will have three hours to complete this exam. No matter when you start the exam, you have to stop at 3:00 PM.
- Notes (except described below), books and calculators/laptops/phones are not allowed.
- A table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k},\ n^k,\ \exp(k),\ n!,\ \Gamma(x),\ \Phi(x),$$

  where $\Phi(x)$ is the CDF of the standard normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.
- You may use 2 handwritten sides of a 8.5" by 11" page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.
  Note: **The total number of points is 70.**
- You **must show all of your work**, unless otherwise stated. A correct answer without explanation will receive only minimal or no partial credit. Note that the "easier" a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

PLEASE CHECK THAT YOU HAVE PAGES 1 – 9, THE DISTRIBUTION TABLE, AND THE MATH FACTS

Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

Part I: For each of the following, simply provide the answer. It is not required that any derivation or justification be provided. Each of these is worth three points.

1. Suppose that A and B are disjoint events. Which of the following is always true?

   Choose One:

   (a) $P(A|B) = P(A)$

   (b) A and B are independent

   (c) $P(A|B) = 0$

   (d) $P(A|B) \geq P(B)$

2. Suppose that there are n stocks in a particular sector. Let X equal the number of these stocks whose price will go up on the next trading day. If I assume that X has the binomial(n, p) distribution, where $0 < p < 1$, then I am assuming that ...

   Choose One:

   (a) ... the probability of at least one stock going up is p.

   (b) ... the stocks are independent in the sense that given that one stock in the sector has gone up, the probability of another stock in that sector going up is still p.

   (c) ... exactly np of the stocks will go up.

   (d) ... $P(X = np) \geq P(X = k)$ for any k.

3. Suppose that X has the Exponential(λ) distribution. Derive the pdf for $Y = X^2$.

4. Suppose that the length of time between "jumps" is well-modelled by the Exponential(λ) distribution. It has been exactly 10.4 hours since the last jump. Which of the following is true? Assume that λ is in units of hour^{-1}.

   Choose One:

   (a) The remaining time until the next jump has the Exponential(λ) distribution.

   (b) The expected time until the next jump is $1/\lambda - 10.4$ hours.

   (c) The total time that will elapse from the previous jump until the next jump will have the Pareto$(\alpha = 1, \lambda)$ distribution.

   (d) The distribution of the remaining time until the next jump depends on how long it has been since the last jump.

5. Suppose that the joint pdf of (X, Y), denoted f_{XY}, is such that $f_{XY}(x, y) > 0$ if and only if $0 < x < 1$ and $0 < y < x$. Which of the following must be true?

   Choose One:

   (a) The marginal density for X, denoted f_X, is such that $f_X(x) > 0$ if and only if $0 < x < 1$.

   (b) Suppose that $0 < y < 1$. The conditional density for X given Y, denoted $f_{X\,|Y}$, is such that $f_{X\,|Y}(x\,|y) > 0$ if and only if $0 < x < 1$.

   (c) The random variables X and Y are independent.

   (d) Without further information, it is impossible to claim that any of (a), (b) or (c) are always true.

6. Suppose that Z_1 has the standard normal distribution, and that U is a random variable with pmf

$$f_U(u) = \begin{cases} 0.5, & \text{if } u = -1, 1 \\ 0, & \text{otherwise} \end{cases}$$

   Assume that U is independent of Z_1. Let $Z_2 = UZ_1$. What is the covariance between Z_1 and Z_2?

7. Suppose that (X, Y) is bivariate normal, and that $E(X) = E(Y) = 2$. What can be said about the random variable $U = 5X + 3Y$?

   Choose One:

   (a) U definitely has the normal distribution, but nothing else can be said about the mean and variance of U, because $\text{Cov}(X, Y)$ is not given.

   (b) U definitely has the normal distribution, and we know that $E(U) = 16$ and $V(U) = 68$.

   (c) U definitely has the normal distribution, and we know that $E(U) = 16$ but the variance of U is unknown, because $\text{Cov}(X, Y)$ is not given.

   (d) We know that $E(U) = 16$, but nothing else can be said about the distribution of U without further information.

Part II: Show your work for these questions, partial credit will be given when appropriate. **Be clear in your steps, and justify them.**

1. Let random variable X have the *chi-squared distribution with k degrees of freedom*, meaning that it has pdf

$$f_X(x) = \begin{cases} \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-x/2}, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

where k is a positive integer. The moment generating function for this distribution is

$$M_X(t) = (1 - 2t)^{-k/2} \quad \text{for } t < 0.5.$$

(a) Show that X has the Gamma distribution. What are the parameters (in terms of k)? Be clear in how you are justifying your conclusion. **(3)**

(b) What are $E(X)$ and $E(X^2)$? **(3)**

(c) Suppose that c is a positive constant. Does cX still have the chi-squared distribution? Remember to justify your response. **(2)**

2. Suppose that (X, Y) has joint pdf

$$f_{(X,Y)}(x, y) = \begin{cases} \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta)}\, x^{\alpha-1}\, y^{\beta-1}, & \text{if } x > 0,\ y > 0, \text{ and } x + y < 1 \\ 0, & \text{otherwise} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.
Note: Remember that $\Gamma(z + 1) = z\Gamma(z)$ for any $z > 0$.

(a) Show that the (marginal) distribution of X is Beta$(\alpha, \beta + 1)$. **(3)**

(b) What is $E(Y)$? **(3)**

(c) What is the covariance between X and Y? (**3**)

(d) Derive the conditional distribution of Y given X. In other words, what is $f_{Y\,|X}$? **(3)**

3. Suppose that there are four envelopes – two are green and two are red. There are also four cards – two are green and two are red. The cards are randomly assigned to the envelopes in such a way that each card is equally likely to end up in each of the four envelopes, but each envelope has exactly one card in it. What is the probability that all four envelopes end up with cards of matching colors? **(3)**
   Hint: Imagine writing a "1" on one red card, and a "2" on another red card. Let A_1 be the event that the "1" card ends up in a red envelope, and let A_2 be the event that the "2" card ends up in a red envelope. How is the event of interest related to these two events?

4. There are n "experts" that each day predict whether or not a particular stock will go up or down. In reality, each of them is flipping a fair coin to make their "prediction." There is a fixed period of m trading days during which their performance will be monitored. What is the probability that there is at least one of the experts that predicts correctly every day of the m trading days? **(3)**

5. Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d., each with the Exponential(λ) distribution. You can assume that $n > 50$.

   (a) Show that $X_{(1)}$, which is the minimum of the X_i has the Exponential(β) distribution. What is the parameter β as a function of λ? **(3)**

   (b) Approximate the probability that $\overline{X}$ is larger than c. As usual, $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. **(3)**

6. Suppose that $X_1, X_2, \ldots X_n$ are i.i.d. random variables, each with the Geometric(p) distribution. Let $Y = \sum_{i=1}^{n} X_i$.

   (a) The table on the last page of the exam includes the *negative binomial*(r, p) *distribution*. Show that Y has the negative binomial distribution with $r = n$. **(3)**

   (b) Which of the following is an appropriate practical interpretation of Y? **(3)**
   Choose One:

      i. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of failures before seeing the n^{th} success.

      ii. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of trials before seeing the n^{th} success.

      iii. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of failures in the first n trials.

      iv. Consider a sequence of independent Bernoulli trials with probability of success p on each. Y equals the number of successes in the first n trials.

7. Suppose that $X_1, X_2, \ldots, X_n$ are the times between successive events in a Poisson process with rate λ. Thus, $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ). A good estimator for the probability that there will be zero events during a period of length one time unit is $g(\overline{X}) = \exp(-1/\overline{X})$, where, as usual, $\overline{X}$ is the sample mean of the X_i.

   Use the Delta method to approximate the distribution of $g(\overline{X})$. You can assume that n is larger than 50. **(3)**

   Note: Your expressions for the mean and variance of the normal distribution should only be in terms of λ and n.

8. We saw in lecture that for a Poisson process with rate λ, the number of events during a time interval of length t has the Poisson(λt) distribution. (Of course, this statement is based on t and $1/\lambda$ being in the same units.)

   (a) Suppose that calls arrive as a Poisson process with rate $\lambda = 10.5$ per hour. What is the probability exactly one call will arrive during a chosen ten minute period? **(3)**

   (b) Imagine that one is not comfortable with assuming that the Poisson process is a reasonable model for calls arriving into the call center. Instead, we want to imagine that the rate of calls varies from hour to hour in an unpredictable manner.

   In particular, we will model the arrival of calls as follows. For each hour period, i.e., from 1:00 to 2:00, from 2:00 to 3:00, and so forth, the rate at which calls arrive follows the Gamma(α, β) distribution. The rate is still in units of hour^{-1}. The rate is independent from hour to the next, and the number of calls is independent from one hour to the next. To be clear: Although it is randomly chosen, the rate is constant during the one-hour period.

   What is the expected number of calls during the hour 2:00 to 3:00 PM? **(3)**

   (c) Reconsider the setup described in part (b) above. Does the number of calls that arrives during the hour 2:00 to 3:00 PM have the Poisson distribution? **(2)**

Useful Math Facts

1. **Results Regarding e:** For any number a,

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \qquad \text{and} \qquad e^a = \lim_{n \to \infty} \left(1 + \frac{a}{n}\right)^n$$

2. **Geometric Series:** If $|r| < 1$ and a is an integer,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}$$

3. **The Binomial Theorem:** For $k = 0, 1, 2, \ldots$,

$$(\lambda_1 + \lambda_2)^k = \sum_{i=0}^{k} \binom{k}{i} \lambda_1^i \lambda_2^{k-i}$$

4. **The Gamma Function:**

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x}\, dx$$

For any t, it holds that $\Gamma(t+1) = t\Gamma(t)$, and for $n = 1, 2, \ldots$, $\Gamma(n) = (n-1)!$. Also, $\Gamma(0.5) = \sqrt{\pi}$.

5. **Definite Integrals:**

$$\int_0^{\infty} e^{-ax}\, dx = \frac{1}{a}, \qquad \int_0^{\infty} x e^{-ax}\, dx = \frac{1}{a^2}, \qquad \int_0^{\infty} x^m e^{-ax}\, dx = \frac{\Gamma(m+1)}{a^{m+1}}$$

$$\int_0^{\infty} e^{-ax^2}\, dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}, \qquad \int_0^{\infty} x e^{-ax^2}\, dx = \frac{1}{2a}, \qquad \int_0^{\infty} x^m e^{-ax^2}\, dx = \frac{\Gamma((m+1)/2)}{2a^{(m+1)/2}}$$

$$\int_0^{1} t^{x-1}(1-t)^{y-1}\, dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \text{The } \textbf{Beta Function}$$

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, $-\infty < x < \infty,\ \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right)$, $0 < x < \infty,\ \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, $0 < x < \infty,\ \alpha, \beta > 0$	$\left(\frac{\beta}{\beta - t}\right)^\alpha, \quad t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x}$, $0 < x < \infty,\ \beta > 0$	$\frac{\beta}{\beta - t}, \quad t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$, $0 < x < 1,\ \alpha, \beta > 0$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a}$, $a \le x \le b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, \quad t \neq 0$ $1, \quad t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x(1-p)^{1-x}$, $x = 0, 1,\ 0 \le p \le 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, \ldots, n,\ 0 \le p \le 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r (1-p)^x$, $x = 0, 1, \ldots,\ r \in \{1, 2, \ldots\},\ 0 \le p \le 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p$, $x = 0, 1, \ldots.\ 0 \le p \le 1$	$\frac{p}{1-(1-p)e^t}, \quad t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda}\,\lambda^x}{x!}$, $x = 0, 1, \ldots,\ \lambda > 0$	$e^{\lambda(e^t - 1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x}\binom{B}{n-x}}{\binom{A+B}{n}}$, $x = 0, 1, \ldots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$

Table 1: Summary of some common distributions in probability and statistics.